**Set 20: Stock Analysis**

**AIM**

To analyze stock prices of three companies using R programming by creating a line chart, calculating daily percentage changes, generating a bar chart, and displaying the stock data in a table.

**ALGORITHM**

1. Create a dataset containing dates and stock prices for Stock A, Stock B, and Stock C.
2. Load the ggplot2 and dplyr libraries.
3. Convert the dataset into long format for plotting.
4. Draw a line chart to visualize stock price trends over time.
5. Calculate the daily percentage change in Stock A prices.
6. Create a bar chart showing the daily percentage changes.
7. Display the stock prices and percentage changes in a table.
8. Interpret the stock price movements.

**R PROGRAM**

```
library(ggplot2)
library(dplyr)
library(tidyr)

stocks <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03")),
  Stock_A = c(100, 105, 110),
  Stock_B = c(150, 152, 148),
  Stock_C = c(120, 118, 122)
)

long_stocks <- pivot_longer(
  stocks, cols = starts_with("Stock"),
  names_to = "Company", values_to = "Price"
)
```

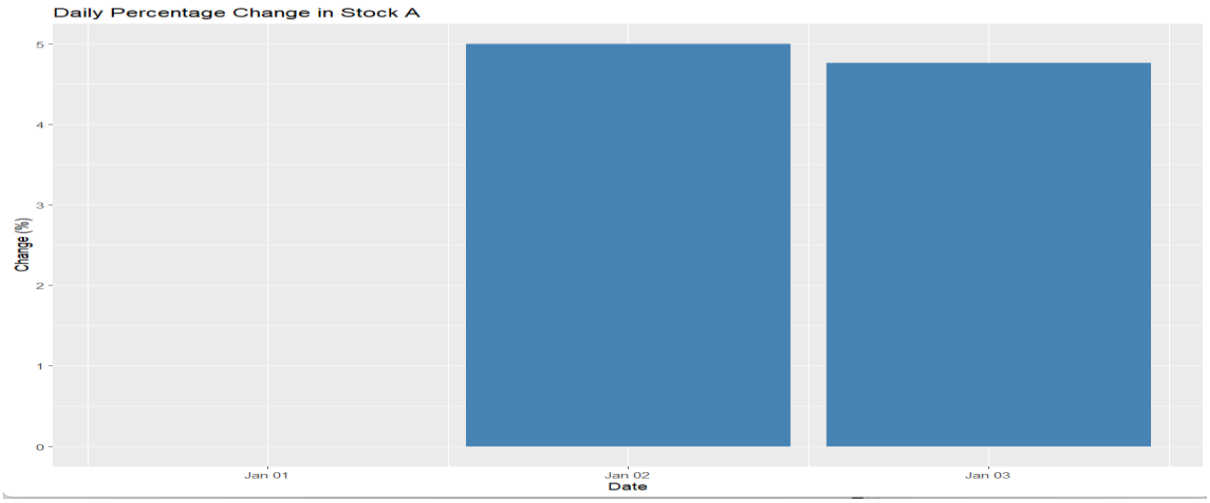
```
ggplot(long_stocks, aes(Date, Price, color = Company)) +
  geom_line(linewidth = 1) +
  geom_point() +
  labs(title = "Stock Price Trends",
       x = "Date", y = "Stock Price")

stocks <- stocks %>%
  mutate(Change = (Stock_A / lag(Stock_A) - 1) * 100)

ggplot(stocks, aes(Date, Change)) +
  geom_col(fill = "steelblue") +
  labs(title = "Daily Percentage Change in Stock A",
       x = "Date", y = "Change (%)")

print(stocks)
```

**OUTPUT:**




**RESULT**

The line chart displays the stock price trends of the three companies. The bar chart shows the daily percentage change in Stock A, and the table presents stock prices and the calculated percentage changes. Stock A increased from 100 to 110 over the three days.